

Securing Kubernetes Cluster with Kubescape



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Date: 14 Aug 2026
Venue: Almaty, Kazakhstan

whoami

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ioka



Yellow



Қазақстандағы
бөлшек сауда желісі
№1

200+ филиалов

открыто за 15 лет в четырнадцати городах Казахстана

40 млрд тенге

выплатила компания «Magnum Cash&Carry» за 15 лет в бюджет

4 млн

человек ежегодно совершают покупки в «Magnum»

1100 и более

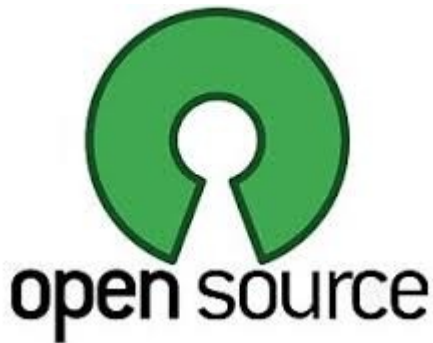
поставщиков работают с сетью, из них более 700 – поставщики товаров казахстанского производства

50 %

всех товаров в сети «Magnum» произведены в Казахстане

What we needed?

- Alternative for Trivy scanner (Trivy Security incident 2026-03-19 #10425)
- Uses Gype engine under the hood
- Open source tool
- Able to scan container images, static manifest files and Kubernetes environment
- Easy integration with CI/CD pipelines
- Easy integration with ArgoCD



What is Kubescape?

Created & maintained by ARMO



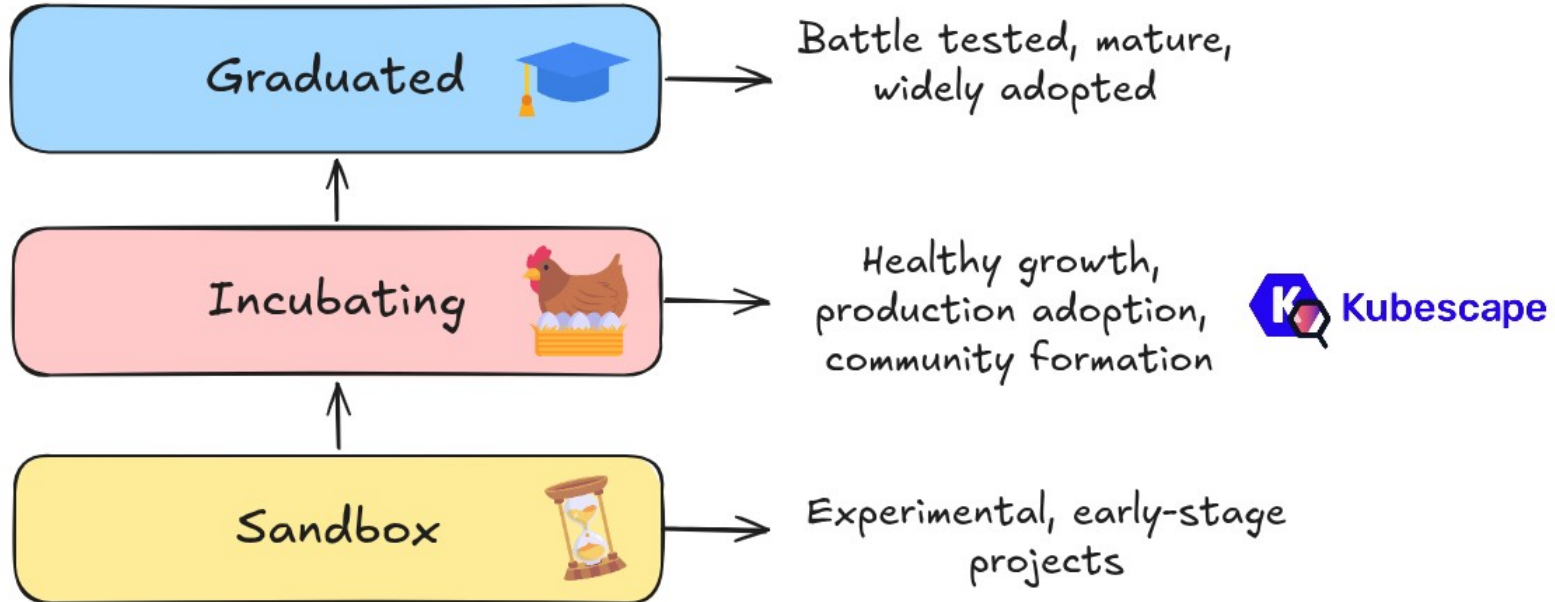
ARMO is an Israeli cybersecurity company focused on Kubernetes security

Donated Kubescape to CNCF to ensure vendor-neutral, community-driven development

ARMO remains the primary contributor and offers a commercial SaaS platform on top of the open-source core

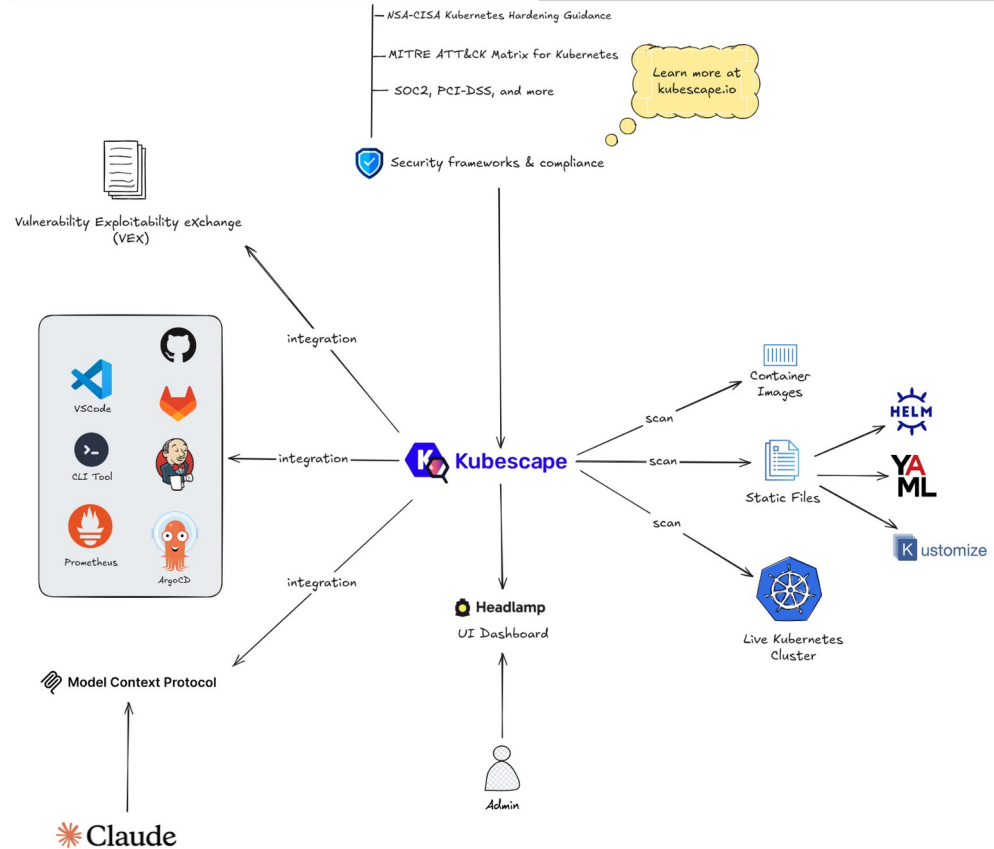
The screenshot shows the Kubescape project page on the CNCF Landscape website. The page header includes the Kubescape logo, CNCF and INCUBATING badges, the year 2022, and the URL <https://landscape.cncf.io/>. Below the header, there are tabs for 'Provisioning' and 'Security & Compliance'. The main content area is titled 'Maturity' and shows a timeline with three stages: '2022-12-13 SANDBOX', '2025-01-13 INCUBATING' (highlighted with a red box), and 'GRADUATED'. Below the maturity timeline, there is a 'Repositories' section with a dropdown menu showing 'kubescape/kubescape (primary)'. Below the dropdown, the GitHub repository URL <https://github.com/kubescape/kubescape> is highlighted with a red box. To the right of the URL are buttons for 'PRIMARY', 'Apache License 2.0', 'good first issues', and '2 open'.

CNCF Maturity Levels



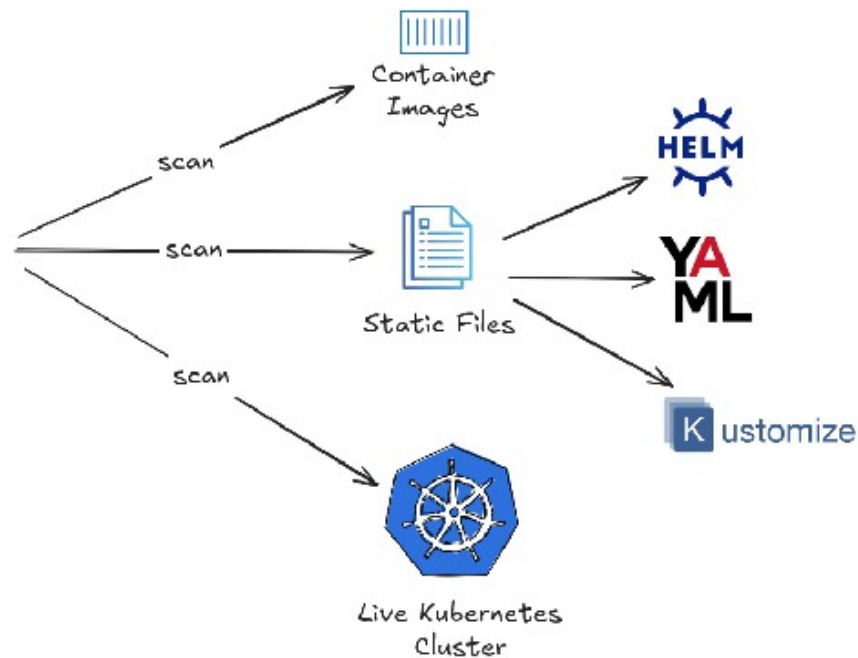
Kubescape Features

- Scans: Container Images, Static Files and Live K8s Cluster
- Scans against the most popular security frameworks and custom rules
- Integrates with ArgoCD, GitLab, Prometheus, and etc.
- Able to provision own MCP server with AI tool connected
- Able to provision own UI dashboard with Headlamp
- Able to generate VEX formatted documents for security teams
- Learn more at kubescape.io



Kubescape Features

Scans: Container Images, Static Files and Live K8s Cluster



Kubescape Features

Scans against the most popular security frameworks and custom rules

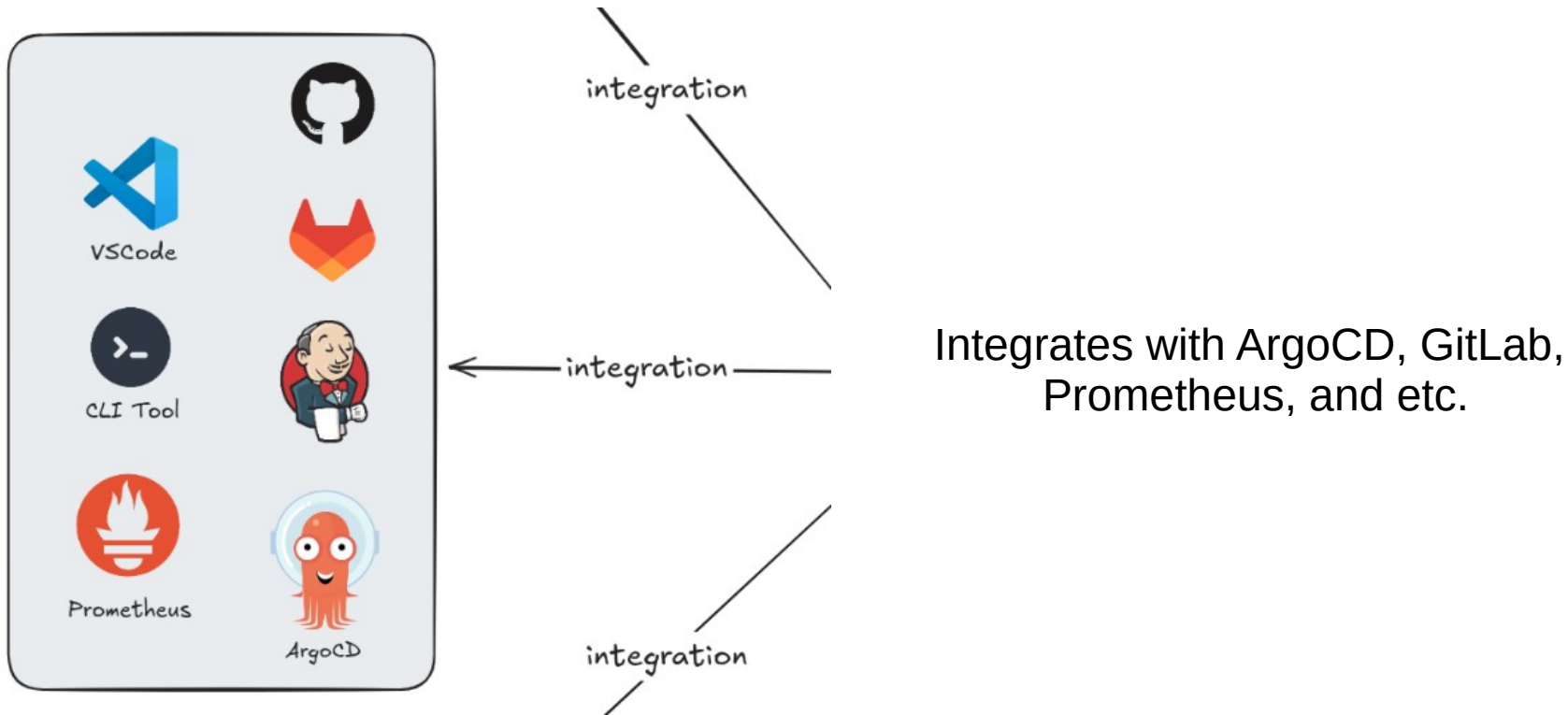
- NSA-CISA Kubernetes Hardening Guidance
- MITRE ATT&CK Matrix for Kubernetes
- SOC2, PCI-DSS, and more



Security frameworks & compliance

Learn more at
kubescape.io

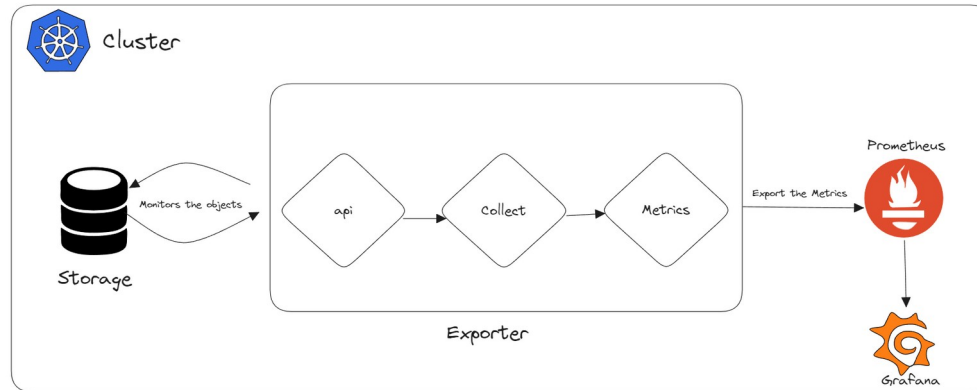
Kubescape Features



Kubescape Features

Kubescape Prometheus metrics

- `kubescape_vulnerabilities_total_cluster_critical` - Total critical CVEs in the cluster
- `kubescape_vulnerabilities_total_namespace_high` - Total high CVEs by namespace
- `kubescape_vulnerabilities_total_workload_critical` - Total critical CVEs by workload
- Node-agent metrics such as `node_agent_alert_counter` when runtime detection is enabled



Kubescape Features

Able to provision own MCP server
with AI tool connected

 Model Context Protocol



 Claude

Kubescape Features

Able to provision own UI dashboard with Headlamp

← → ↻ http://localhost:4466/#/c/main/kubescape/vulnerabilities

Headlamp Search Press / to search

Home Clusters main Map Workloads Storage Network Gateway (beta) Security Configuration Custom Resources **Kubescape** Compliance Vulnerabilities Network Policies Policy Playground Runtime Detection Exceptions Frameworks + Create

Vulnerabilities

5 scans

CVEs Resources Images

5 workload scans, 664 CVE issues

Relevant Fixed

Severity	CVE ID	CVSS	Component	Relevant	Fixed	Images	Workloads	Description
Critical	CVE-2025-68121	10	stdlib go1.15	Yes	Yes	3	4	During session resumption in crypto/tls, if the underlying Config has its ClientCAs or RootCAs fields mutated between the initial handshake and the resumed handshake, the resumed handshake may succeed when it should have failed. This may happen when a user...
Critical	CVE-2021-3711	9.8	libssl1.1 1.1.1d-0+deb10u3 openssl 1.1.1d-0+deb10u3	No	Yes	3	4	In order to decrypt SM2 encrypted data an application is expected to call the API function EVP_PKEY_decrypt(). Typically an application will call this function on twice. The first time, on entry, the "out" parameter can be NULL and, on exit, the "outlen" para...
Critical	CVE-2021-25216	9.8	bind9-host 1:9.11.5.P4+dfsg-5.1+deb10u2 dnstools 1:9.11.5.P4+dfsg-5.1+deb10u2 libbind9-161 1:9.11.5.P4+dfsg-5.1+deb10u2 libdns1104 1:9.11.5.P4+dfsg-5.1+deb10u2 liblrs161 1:9.11.5.P4+dfsg-5.1+deb10u2 libisc1100 1:9.11.5.P4+dfsg-5.1+deb10u2 libisc161 1:9.11.5.P4+dfsg-5.1+deb10u2 libisc163 1:9.11.5.P4+dfsg-5.1+deb10u2 liblwres161 1:9.11.5.P4+dfsg-5.1+deb10u2	No	Yes	3	4	In BIND 9.5.0 → 9.11.29, 9.12.0 → 9.16.13, and versions BIND 9.11.3-S1 → 9.11.29-S1 and 9.16.8-S1 → 9.16.13-S1 of BIND Supported Preview Edition, as well as release versions 9.17.0 → 9.17.1 of the BIND 9.17 development branch, BIND servers are vulnera...



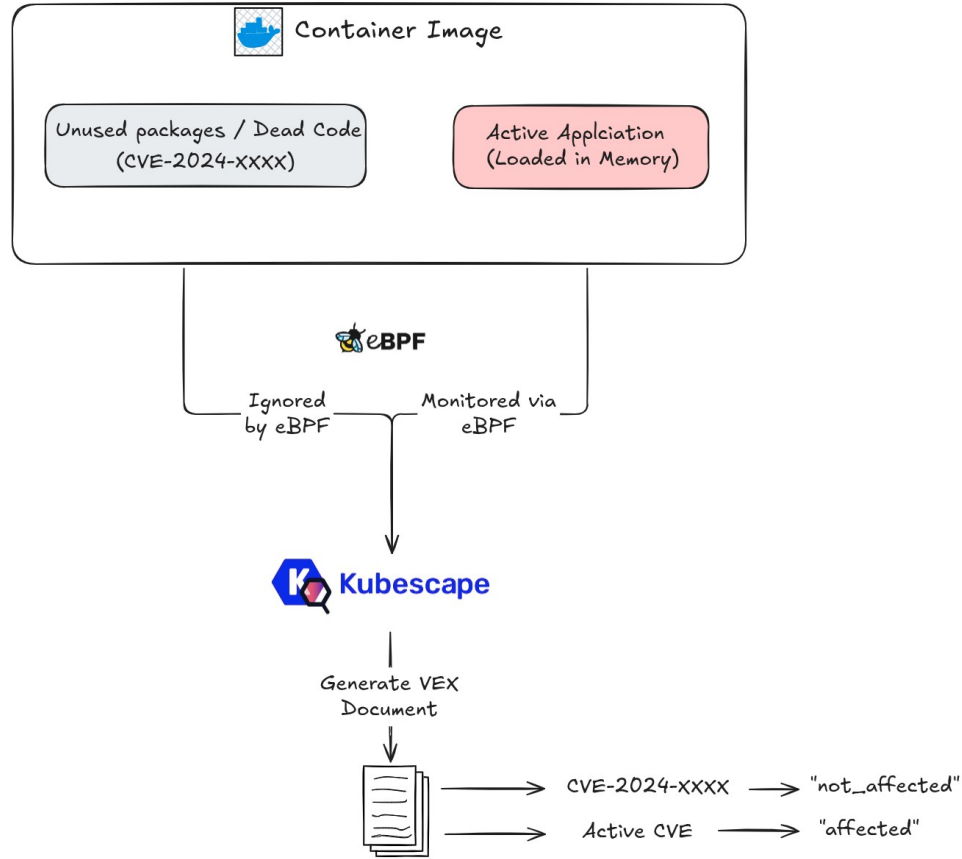
UI Dashboard



Admin

Kubescape Features

Able to generate VEX
formatted documents
for security teams



Scan Cluster with Kubescape CLI

```
Kubescape Basic Commands

# Scan cluster against default frameworks (MITRE + NSA)
kubescape scan

# List available frameworks
kubescape list frameworks
kubescape list controls

# Scan against a specific framework
kubescape scan framework nsa
kubescape scan framework mitre

# Download a specific framework
kubescape download framework nsa

# Scan a specific control with details
kubescape scan control C-0260 -v

# Scan local manifest files
kubescape scan nginx-app.yml

# Scan a specific namespace only
kubescape scan --include-namespaces app

# Scan image
kubescape scan image nginx:latest-
```

V4.0.9

Supported frameworks
AllControls
ArmoBest
DevOpsBest
MITRE
NSA
SOC2
cis-aks-t1.2.0
cis-aks-t1.8.0
cis-eks-t1.7.0
cis-eks-t1.8.0
cis-gke-v1.9.0
cis-v1.10.0
cis-v1.12.0

Scan Cluster with Kubescape CLI

Secrets

Control name	Resources	View details
Applications credentials in configuration files	1	\$ kubescape scan control C-0012 -v
Automatic mapping of service account	3	\$ kubescape scan control C-0034 -v

Network

Control name	Resources	View details
Cluster internal networking	2	\$ kubescape scan control C-0054 -v
Container hostPort	1	\$ kubescape scan control C-0044 -v
HostNetwork access	1	\$ kubescape scan control C-0041 -v
Ingress and Egress blocked	3	\$ kubescape scan control C-0030 -v
Missing network policy	4	\$ kubescape scan control C-0260 -v

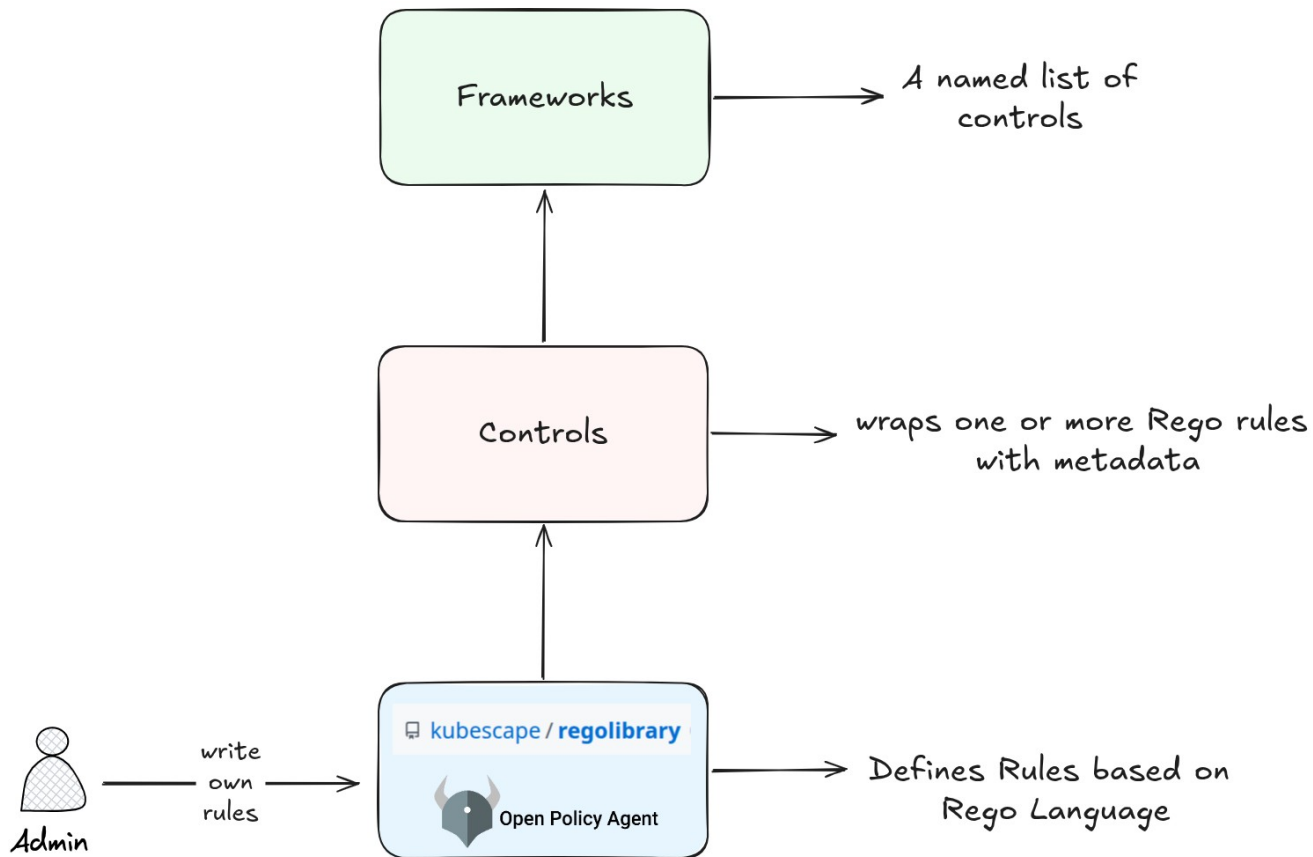
Workload

Control name	Resources	View details
CVE-2021-25741 - Using symlink for arbitrary host file system access.	0	\$ kubescape scan control C-0058 -v
CVE-2021-25742-nginx-ingress-snippet-annotation-vulnerability	0	\$ kubescape scan control C-0059 -v
Exposed sensitive interfaces	0	\$ kubescape scan control C-0021 -v
Kubernetes CronJob	0	\$ kubescape scan control C-0026 -v
Mount service principal	0	\$ kubescape scan control C-0020 -v
SSH server running inside container	0	\$ kubescape scan control C-0042 -v

Supply chain

Control name	Resources	View details
Anonymous access enabled	1	\$ kubescape scan control C-0262 -v
Authenticated user has sensitive permissions	0	\$ kubescape scan control C-0265 -v

Create Your Own Baseline Framework



Create Your Own Baseline Framework

kubescape / regolibrary Public

Notifications Fork 70 Star 131

[Code](#) [Issues 10](#) [Pull requests 6](#) [Actions](#) [Projects](#) [Security and quality](#) [Insights](#)

Files

master

Go to file

> .github

> attack-tracks

> categories

> controlIdsmigration

> controls

> examples

C-0001-forbiddencontainerreg...

C-0002-execintocontainer.json

C-0004-resourcesmemorylimit...

C-0005-apiserverinsecureporti...

C-0007-datadestruction.json

C-0009-resourcelimits.json

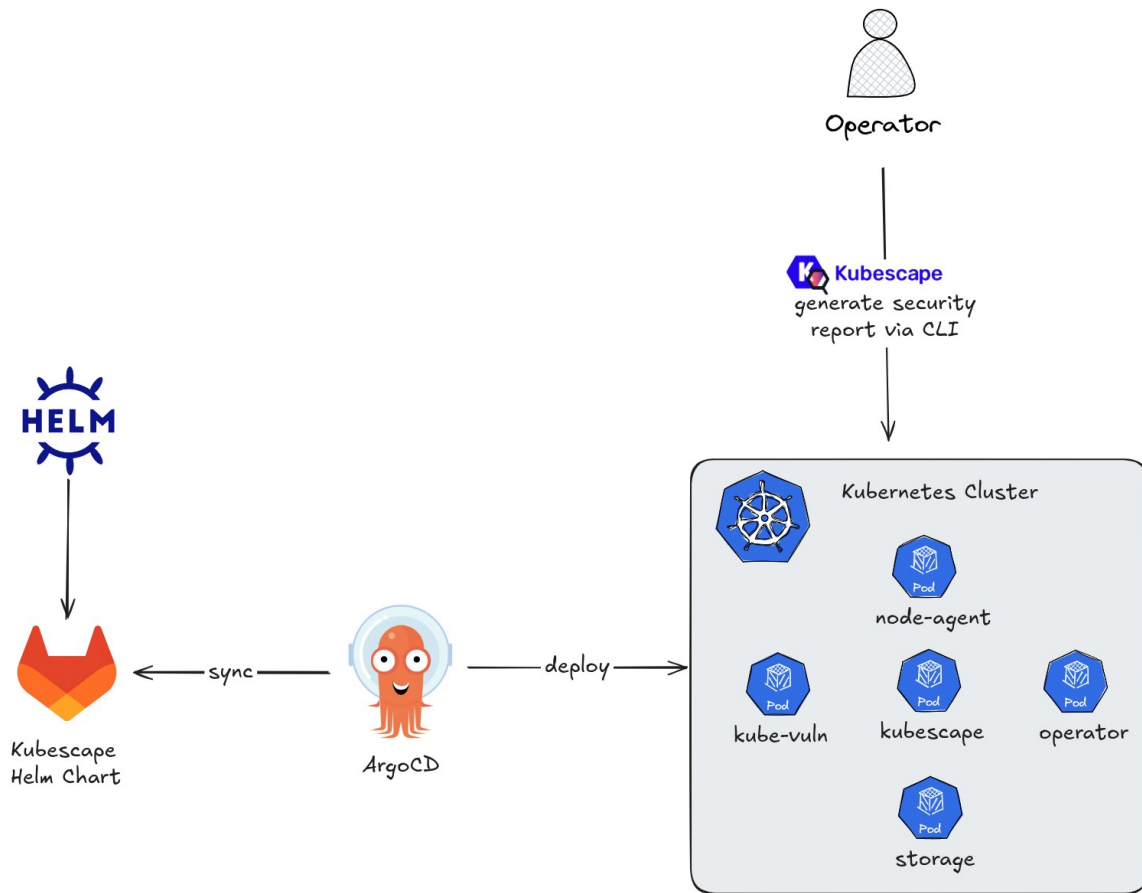
C-0012-applicationscredential...

regolibrary / controls

GeneCodeSavvy feat(rule): improve mismatching selector parity aa566af · last month History

Name	Last commit message	Last commit da...
..		
examples	feat(rules): full CiliumNetworkPolicy support for C-0030, C-0054, C-0260	last month
C-0001-forbiddencontainerregistries.json	Revert "delete C-0001"	2 years ago
C-0002-execintocontainer.json	rename C-0002 to Prevent containers from allowing command execution	2 years ago
C-0004-resourcesmemorylimitandrequest.json	removing armoBuiltin from all controls	3 years ago
C-0005-apiserverinsecureportisenabled.json	removing armoBuiltin from all controls	3 years ago
C-0007-datadestruction.json	rename C-0007 to Roles with delete capabilities	2 years ago
C-0009-resourcelimits.json	split cpu and memory controls (#594)	2 years ago
C-0012-applicationscredentialsinconfigurationfiles.json	removing armoBuiltin from all controls	3 years ago
C-0013-nonrootcontainers.json	clarify remediation	2 years ago

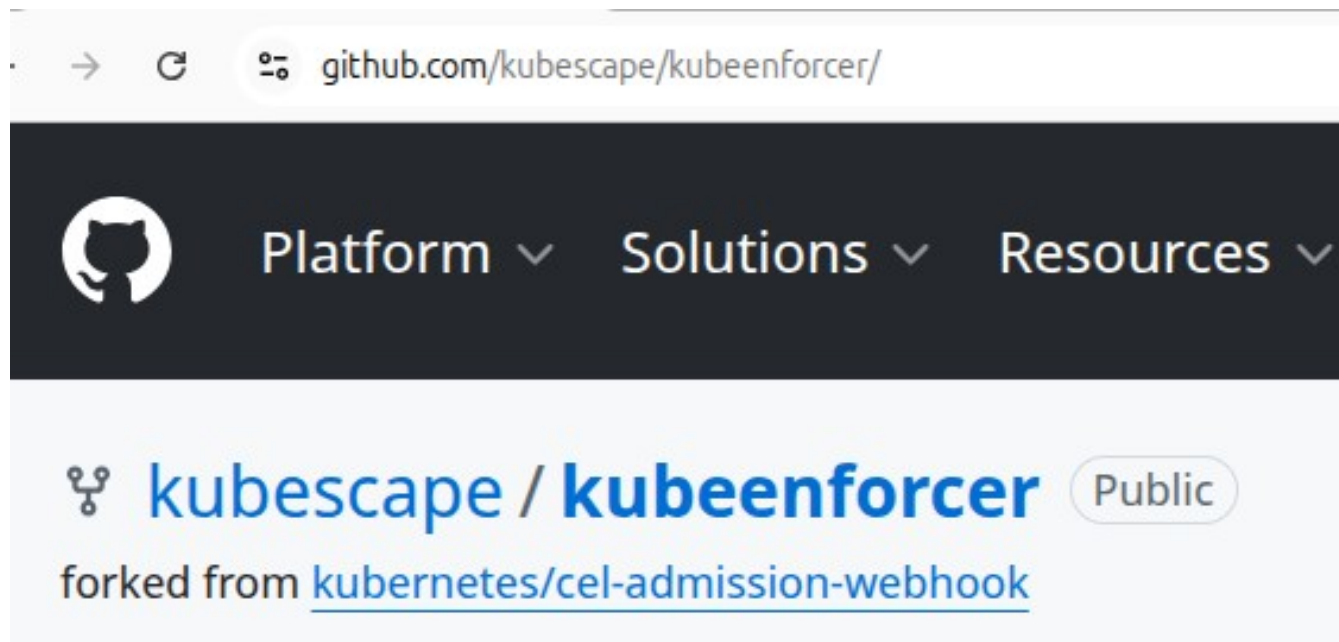
Dynamic Kubescape



Kubescape Helm Values

```
kubescape-operator:  
  clusterName: test-cluster  
  
  excludeNamespaces: "kubescape,kube-system,kube-public,kube-node-lease,kubeconfig,gmp-system,gmp-public,cattle-system"  
  
  capabilities:  
    configurationScan: disable  
    continuousScan: disable  
    nodeScan: disable  
    vulnerabilityScan: disable  
  
  grypeOfflineDB:  
    enabled: true  
  
  persistence:  
    storageClass: longhorn  
  
  kubevuln:  
    tmpDirPersistence:  
      enabled: true  
      storageClassName: longhorn  
      size: 3Gi  
    config:  
      grypeDbPersistence: true
```

Kubeenforcer for Alermanager



Kubeenforcer Configs

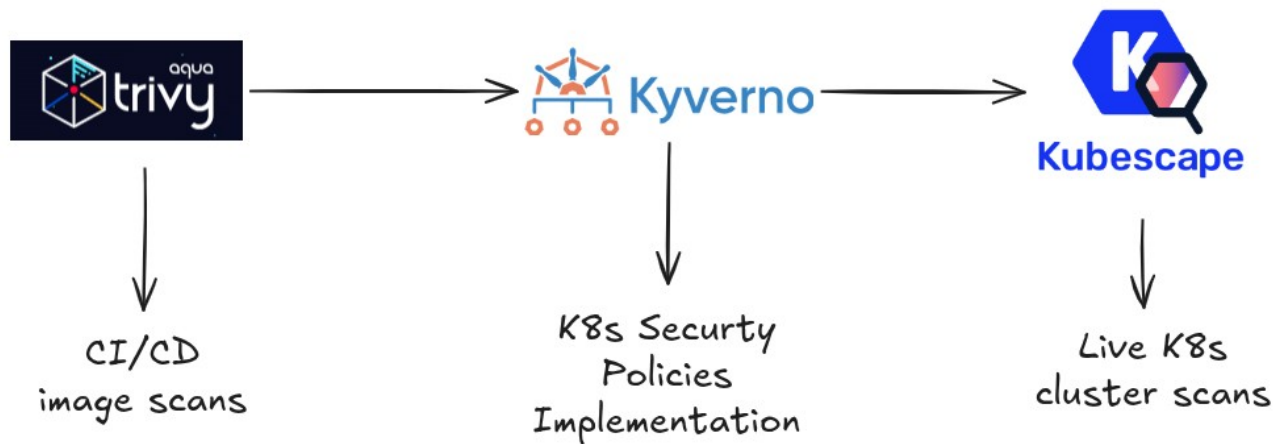
```
kind: ConfigMap
apiVersion: v1
metadata:
  name: alertmanager-config
  namespace: kubescape
data:
  alertmanager.yml: |-
    global:
      smtp_smarthost: smtp-server:port    # SMTP server and port
      smtp_from: sender@email.com        # Sender email address
      smtp_auth_username: ""              # SMTP username
      smtp_auth_password: ""              # SMTP password
      smtp_require_tls: true               # Require TLS encryption

    route:
      group_by: ['alertname']
      receiver: email-notifier

    receivers:
      - name: email-notifier
        email_configs:
          - to: your@email.com
```

What Values Kubescape Provides?

- Covers static and dynamic analysis of your Kubernetes environment and its configurations
- Provides ability to form your own custom security framework aligned with your organization's security policies
- Reduce security noise and prioritize “real” risks by using VEX reports
- It's not a “silver bullet” for security and best it can shine in combination with other security tools like: Trivy (CI, image scans) + Kubescape (live Kubernetes scans)



Demo

<https://bit.ly/cncf-almaty-2026-meetup-kubescape>

